

LAST LIGHT

Technical Art Baseline

Portrait Mobile | Unity | Pixel Art | PM-Reduced P0 Scope



Purpose

Lock the screen reference, render order, sprite dimensions, import rules, and production constraints needed to integrate art without delaying gameplay gates G1-G3.

Based on the Last Light Art Direction Document v1.0 and the approved reduced mobile-readable production plan.

1. Reference Screen and Camera

Reference resolution

Setting	Value
Reference canvas	360 x 640 px
Orientation	Portrait, 9:16
Game-world PPU	16 pixels per unit
Canvas Scaler	Scale With Screen Size
Canvas reference	360 x 640
Canvas Match	0.5
Texture filtering	Point
Pixel Perfect Camera	Enabled
Reference resolution	360 x 640
Assets PPU	16

Camera calculation. At 16 PPU, the 360 x 640 frame represents 22.5 x 40 Unity world units. A conventional orthographic camera matching the full reference height starts at orthographic size 20; the Pixel Perfect Camera remains the final authority.

Screen composition zones

Zone	Reference pixels	Purpose
Top 20%	128 px high	Upcoming hazard warnings and stage progress
Middle 50%	320 px high	Main play space and highest-contrast lanes
Bottom 30%	192 px high	Player focus and touch-safe movement area
Outer edges	Variable	Walls, vignette and world decay

- Do not place important text or choice cards under the player thumb area.
- Keep the upper warning area free of dense props.
- Outer-edge darkening may increase during the run but must not conceal fair telegraphs.

Mobile UI measurements

- Minimum essential text: 18 reference px.
- Choice target width: at least 120 px per choice.
- Minimum center gap: 32 px.
- At 360 px width, two 120 px choices plus a 32 px gap leave approximately 44 px per side.
- Apply device safe-area offsets before positioning the HUD.
- Use a readable sans-serif font; do not rely on a tiny bitmap font.

2. Unity Rendering and Sorting Structure

Use four Sprite Sorting Layers, followed by the UI Canvas:

1. Background
2. World
3. WorldFX
4. ScreenFX
5. UI Canvas

Back-to-front render order

#	Sorting Layer	Order	Object
1	Background	0	background_void

#	Sorting Layer	Order	Object
2	Background	10	background_far_corridor
3	Background	20	background_decay_overlay
4	World	0	level_floor
5	World	10	level_floor_decals
6	World	20	level_borders_back
7	World	30	level_props_back
8	World	40	level_altars
9	World	50	obstacle_bodies
10	World	60	player_group
11	World	70	level_borders_front
12	WorldFX	0	hazard_telegraphs
13	WorldFX	10	altar_selection_fx
14	WorldFX	20	cost_transfer_particles
15	WorldFX	30	damage_particles
16	WorldFX	40	stage_clear_fx
17	WorldFX	50	final_delivery_fx
18	ScreenFX	0	global_palette_tint
19	ScreenFX	10	vignette_or_vision_mask
20	ScreenFX	20	damage_screen_frame
21	ScreenFX	30	scene_transition_fade
22	UI Canvas	100	Persistent HUD
23	UI Canvas	200	Choice cards and labels
24	UI Canvas	300	Critical warnings and result feedback
25	UI Canvas	400	Pause, settings and development debug

Rendering decisions

- Hazard bodies render behind the player so the traveller remains legible during contact.
- Hazard telegraphs render above the entire world, including the player and front borders.
- UI remains above screen effects except during a deliberate full-screen transition.
- Avoid automatic Y-sorting unless gameplay truly requires it; fixed sorting is more predictable for this corridor.
- Reserve order increments of 10 so temporary elements can be inserted later.

3. Internal Player Sprite Order

Place traveller visuals under one Unity SortingGroup:

```
Sorting Layer: World
Order in Layer: 60
```

Relative order	Child
-3	GlowOuter
-2	GlowInner
0	TravellerBase
1	BodyScaleCorrection
2	OverlayChain

Relative order	Child
3	OverlayCrack
4	OverlayCorruptionOrHiddenSenses
5	RelicCore
6	HitFlash

Recommended player prefab

```

PF_Traveller
├── VisualRoot [SortingGroup: World / 60]
│   ├── GlowOuter
│   ├── GlowInner
│   ├── TravellerBase
│   ├── BodyScaleCorrection
│   ├── OverlayChain
│   ├── OverlayCrack
│   ├── OverlayCorruption
│   ├── RelicCore
│   └── HitFlash
├── Collider
├── MovementController
└── StateController

```

- Keep the relic as a separate child sprite so scale, alpha and flicker can respond independently to relic percentage.
- Keep cracks, chains, glow and corruption as independent overlays. Do not create a unique sheet for every sacrifice combination.

Player pivot and frame rules

- Pivot: Bottom Center.
- Every frame uses the same 24 x 24 canvas.
- Every frame shares the same bottom-center pivot and baseline row.
- Preserve the same frame bounds.
- Avoid subpixel movement.
- Do not generate the collider from visible sprite bounds; use a smaller primitive collider.

4. Other Prefab Layer Structures

Altar prefab

```

PF_Altar
├── VisualRoot [SortingGroup: World / 40]
│   ├── Base [relative 0]
│   ├── InsetOrIcon [relative 1]
│   └── IdlePulse [relative 2]
├── ChoiceLabelAnchor
├── TriggerCollider
└── ChoiceController

```

The safety and cost altars share the same outer mass. Differentiate them through inset, icon, pulse and feedback.

Moving blade prefab

```

PF_Hazard_Blade
├── VisualRoot [SortingGroup: World / 50]
│   ├── BladeBody [relative 0]
│   ├── PurpleHub [relative 1]
│   └── ContactFlash [relative 2]
├── Telegraph [WorldFX / 0]

```

```
└─ Collider
└─ MovementController
```

P0 uses one static blade sprite. Rotate or translate it in Unity rather than producing animation frames.

Spike prefab

```
PF_Hazard_Spikes
└─ SpikeModules      [World / 50]
└─ Collider
└─ HazardController
```

Build long spike walls by repeating a 16 x 16 module. Keep colliders simple and independent from visible triangles.

5. P0 Asset Sizes - Mobile-Readable Minimum

These assets are integrated only after G3 is stable.

Asset	Count	Source size	Recommended export
Static traveller	1	24 x 24	chr_traveller_base.png
Relic core	1	8 x 8	fx_relic_core.png
Inner relic glow	1	16 x 16	fx_relic_glow_inner.png
Spike module	1	16 x 16	haz_spike_module.png
Moving blade	1	32 x 32	haz_blade_base.png
Safety altar	1	32 x 48	altar_safe_idle.png
Cost altar	1	32 x 48	altar_cost_idle.png
Cost icons	5	24 x 24 each	Separate PNGs or 120 x 24 sheet
Floor clean	1	16 x 16	env_floor_clean.png
Floor cracked	1	16 x 16	env_floor_cracked.png
Wall edge left	1	16 x 16	env_wall_edge_left.png
Wall edge right	1	16 x 16	env_wall_edge_right.png
Wall cap/corner	1	16 x 16	env_wall_cap.png
Background void strip	1	16 x 32	env_void_strip.png
HUD	1	Unity UI	Avoid bespoke imagery in P0

HUD recommendation

- Build the first HUD from Unity UI primitives.
- Use solid-color bar backgrounds and fills.
- Reuse the 24 x 24 icons.
- Use readable sans-serif text.
- A 1 x 1 white texture may be used for tintable shapes.
- Do not draw elaborate HUD frames before the complete run works.

6. P1 Asset Sizes

Asset	Count	Source size	Implementation
Traveller move	2 frames	24 x 24 each	48 x 24 sheet
Traveller hurt	1 frame	24 x 24	Separate or appended
Crack overlay	1	24 x 24	Same pivot as player
Chain overlay	1	24 x 24	Same pivot as player
Sensory-loss overlay	1	24 x 24	Same pivot as player

Asset	Count	Source size	Implementation
Shadow warning	1	16 x 32	Transparent PNG
Shadow strike	1	16 x 32	Transparent PNG
Ending silhouettes	2	180 x 120 minimum	Separate PNGs
Square particle	1 optional	2 x 2 or 4 x 4	Tint procedurally
Full-screen vignette	1	360 x 640	3-4 stepped bands
Vision mask	1-3 states	360 x 640	Hard pixel steps, no blur

The full art direction allows a four-frame move and two-frame hurt animation, but the approved staging begins with a static traveller and adds minimum motion after P0 integration.

7. P2 and Optional Assets

Asset	Size
Additional move frames	24 x 24 each
Full four-frame move sheet	96 x 24
Two-frame idle flicker	48 x 24
Delivery poses	24 x 24 each
Death/collapse frames	24 x 24 each
Debris props	8 x 8 to 16 x 16
Rune/stain overlays	8 x 8 to 16 x 16
Outer relic glow	Up to 20 x 20
Additional ending	180 x 120 minimum

Ending art may be authored at 360 x 240 and reduced to 180 x 120 with nearest-neighbour scaling, provided the composition remains silhouette-led.

8. Pixel and Alpha Rules

Opaque world sprites

- Use alpha 255 for visible pixels and alpha 0 for transparent pixels.
- Do not use partially transparent edge pixels on traveller, hazards, altars, tiles, icons or props.
- Partial alpha is allowed only for relic glow layers, large screen overlays, vignette, palette fade and selected particles.
- Use hard one-pixel clusters, no anti-aliased edges, no isolated noise pixels and no subpixel animation.

Locked palette

Token	Hex	Reserved use
Void	#0A0D14	Outermost background and deepest shadow
Deep Navy	#121B2D	Corridor and UI dark
Stone	#2A3142	Walls, props and neutral hazard mass
Bone	#D9E1E8	Traveller and readable neutral text
Relic Gold	#F6C85F	Relic and direct relic consequences only
Danger Red	#D94A5A	Direct harm and damage only
Safe Cyan	#5FA9C4	Safety and temporary guidance
Shadow Purple	#75598E	Shadow threat, corruption and sensory cost

- Do not use gold as environment decoration.
- Do not use red for safe choices.

- Purple must never visually overpower gold.

9. Unity Import Preset

Create one preset named:

PRESET_PixelArt_World

Import setting	Value
Texture Type	Sprite (2D and UI)
Sprite Mode	Single or Multiple as appropriate
Pixels Per Unit	16
Filter Mode	Point
Compression	None
Generate Mip Maps	Off
Mesh Type	Full Rect
Extrude Edges	0-1
Pivot	Custom or fixed preset
Max Size	At least source dimensions
Read/Write	Off unless a tool requires it

Sprite Atlas rules

- Padding: 2 px minimum.
- Rotation: Off.
- Tight Packing: Off.
- Filter: Point.
- Compression: None.
- Keep UI and world art in separate atlases.

10. Lighting and Screen Effects

Do not use bloom as the core relic-light solution.

RelicCore 8 x 8 alpha 1.0
 GlowInner 16 x 16 controlled alpha
 GlowOuter up to 20 x 20, optional

Relic value	Visual state
70-100%	Core, inner glow and optional outer glow
30-69%	Core and smaller inner glow
1-29%	Dim core with intermittent flicker
0%	No active gold

- Use one shared global material or overlay for palette desaturation.
- Do not instantiate a different material for every environment sprite.
- Use layered nearest-neighbour glow sprites rather than heavy post-processing.

11. VFX Limits

Effect	Technical limit
Damage flash	0.15-0.25 seconds
Cost selection	0.45-0.70 seconds
Relic loss	0.30-0.50 seconds
Vision mask change	0.40-0.60 seconds
Camera shake	Under 0.15 seconds
Hit stop	Under 0.06 seconds
Particles	8-16 squares per event
Full-screen overlays	One vignette plus bounded feedback
Continuous emission	Relic flicker only

- Use pooled particle systems.
- Avoid full-screen blur.
- Avoid long trails and continuous screen shake.

12. Folder and Naming Structure

```
Assets/  
├── Art/  
│   ├── Generated/  
│   │   └── References/  
│   ├── Source/  
│   │   └── Aseprite/  
│   ├── Production/  
│   │   ├── Characters/  
│   │   ├── Hazards/  
│   │   ├── Environment/  
│   │   ├── Altars/  
│   │   ├── UI/  
│   │   └── VFX/  
│   ├── Atlases/  
│   ├── Materials/  
│   └── Presets/  
├── Prefabs/  
│   ├── Characters/  
│   ├── Hazards/  
│   ├── Environment/  
│   └── UI/  
└── Animations/
```

Naming examples

```
chr_traveller_base.png  
chr_traveller_move.png  
ovr_traveller_crack.png  
ovr_traveller_chain.png  
  
fx_relic_core.png  
fx_relic_glow_inner.png  
  
haz_spike_module.png  
haz_blade_base.png  
haz_shadow_warning.png  
haz_shadow_strike.png
```



```

altar_safe_idle.png
altar_cost_idle.png

env_floor_clean.png
env_wall_edge_left.png

ui_icon_health.png
ui_icon_light.png

PF_Traveller
PF_Hazard_Blade
PF_Altar_Safe

AN_Traveller_Move
AN_Traveller_Hurt

MAT_PaletteFade

```

Keep editable source files beside, or clearly mapped to, their exports. AI-generated references remain outside the production sprite folder until manually redrawn and approved.

13. Development Rules That Save Time

1. G1-G3 use primitives. Do not wait for sprites before movement, collisions, hazards, choices and the ending work.
2. Integrate one vertical slice first: one traveller, one spike, one altar and one icon. Test on a phone before producing the rest.
3. Never derive gameplay from artwork. Colliders, movement paths and choice zones are gameplay objects; sprites are replaceable visual children.
4. Use one SortingGroup per moving object so child overlays do not separate when objects cross other world elements.
5. Use shared materials. Palette fading and relic-state tinting must not create per-object material instances.
6. Keep all world positions pixel-snapped. At 16 PPU, one source pixel equals 0.0625 Unity units.
7. Do not resize sprites with arbitrary transforms. Prefer 1x, 2x or another controlled scale. The enlarged-body sacrifice is the exception and may require an outline-correction overlay.
8. Test with effects disabled. Traveller, relic, hazards and altars must remain readable without glow, particles or shake.
9. Test at 25% desktop zoom and on the target phone.
10. Do not create new biomes or full replacement character sheets. Variation comes from palette shifts, overlays, prop density and light loss.

Production Gate Summary

Gate	Allowed art state	Exit condition
G1 - Core movement	Primitive player, walls and collision shapes only	Movement and camera stable on device
G2 - Core loop	Unity primitives; text labels permitted	Hazard timing, choice collision and state changes work
G3 - End-to-end	All placeholders acceptable	Full run completes on target phone without art dependency
Post-G3 - P0 integration	Replace only mobile-critical placeholders	P0 integrated without readability or schedule regression

Art-director rule: No visual feature is allowed to delay the first complete playable build. One coherent visual language with a small asset set is more valuable than many inconsistent assets.